

DAY 4 Milestone: **OVERVIEW :**

Analytics Output & Dashboard Visualization

This milestone focuses on visualizing real-time dam sensor data and machine learning prediction results using a dashboard. The goal is to provide clear insights for monitoring dam conditions and identifying risk levels.

Objective :

To visualize real-time dam data and ML predictions using interactive dashboards for better analysis, monitoring, and early risk detection.

Data Source :

The data used in this milestone is generated from **Milestone 3 (Live Data Simulation & Model Integration)**.

Dataset includes:

- Timestamp
- Rainfall
- Water Level
- Pressure
- Predicted Water Level
- Risk Level

Tools Used :

- **Power BI** – for dashboard visualization
- **CSV Dataset (live_predictions.csv)** – for input data

Dashboard Overview :

The dashboard provides a clear and real-time view of dam conditions using multiple visualizations. It helps in understanding trends, predictions, and risk levels in an easy and interactive manner.

Visualizations Created :

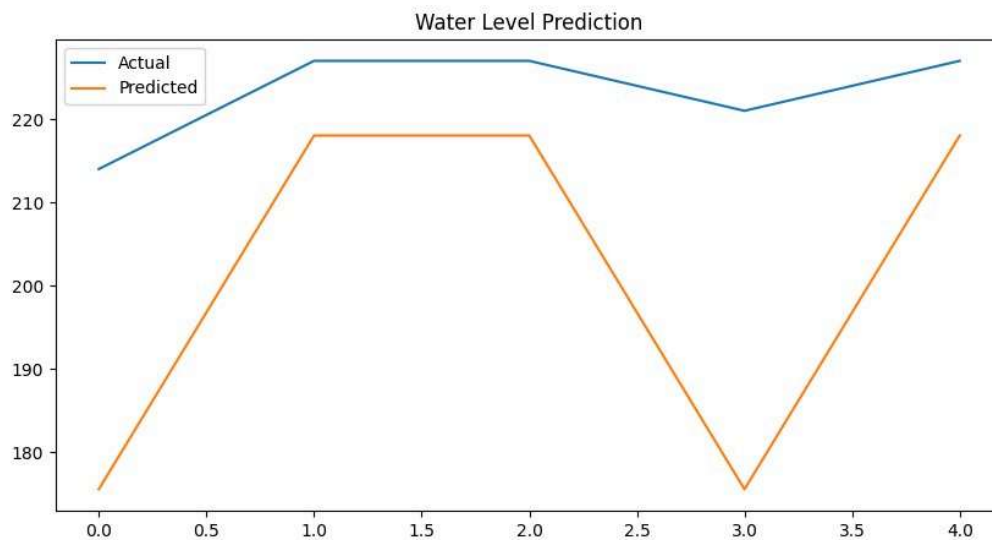
1. Water Level Trend (Line Chart)

- Displays actual water level and predicted water level over time
- Helps in analyzing prediction accuracy

Insight:

Predicted values closely follow actual values, showing good model performance.

Output :

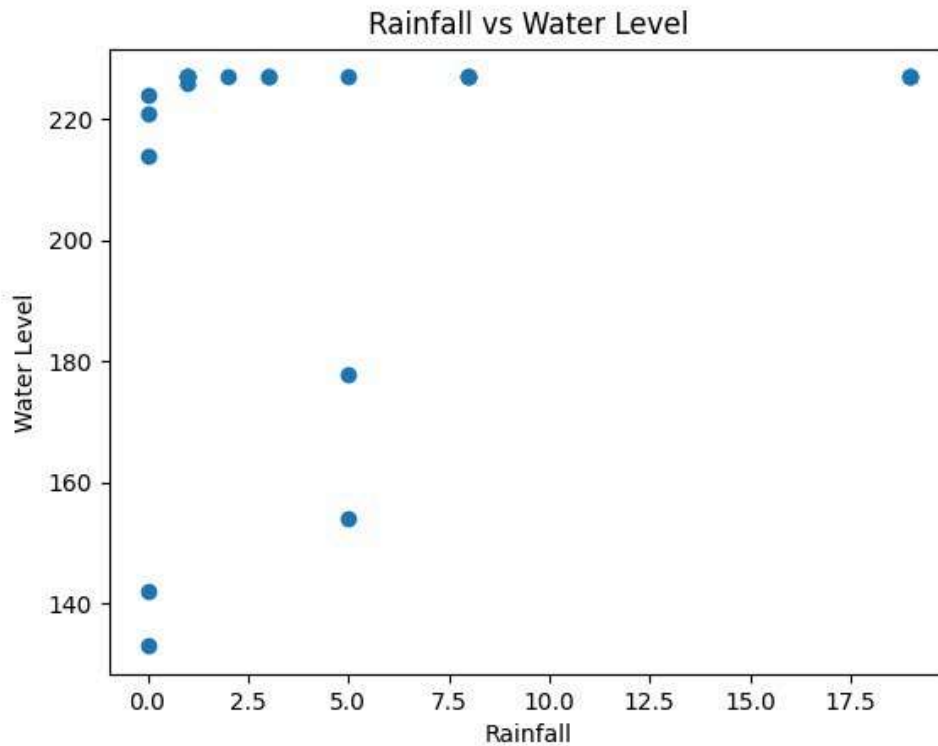


2. Rainfall vs Water Level (Scatter Chart)

- Shows the relationship between rainfall and water level
- Helps in understanding how rainfall affects reservoir level

Insight:

As rainfall increases, water level also increases.

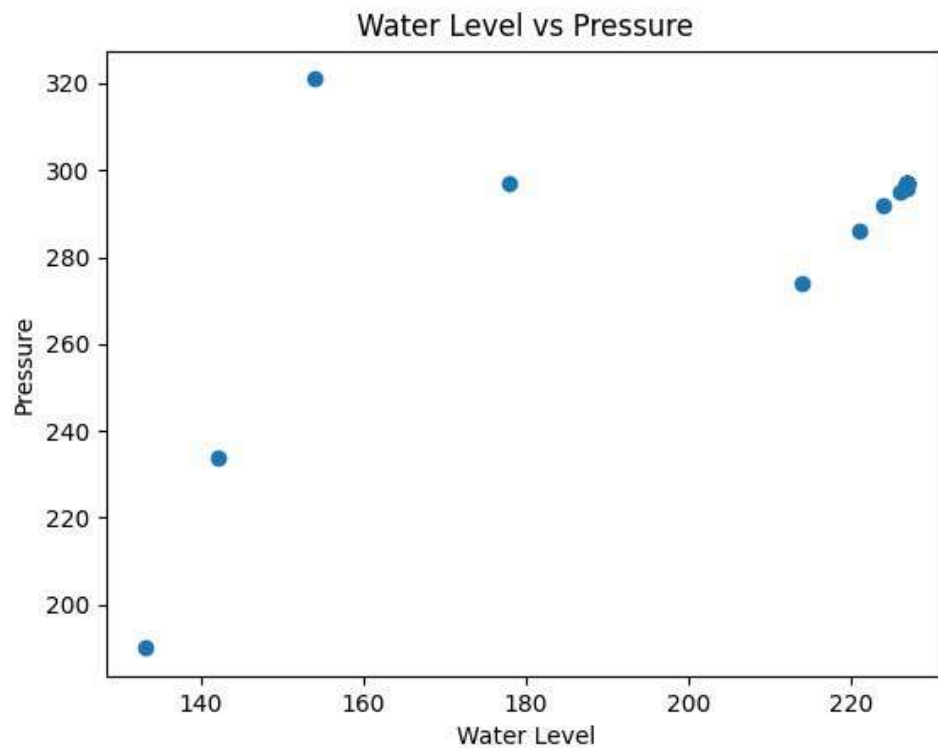
Output :**3. Pressure vs Water Level (Scatter Chart)**

- Shows how pressure changes with water level
- Useful for analyzing structural stress on the dam

Insight:

Pressure increases as water level increases.

Output :



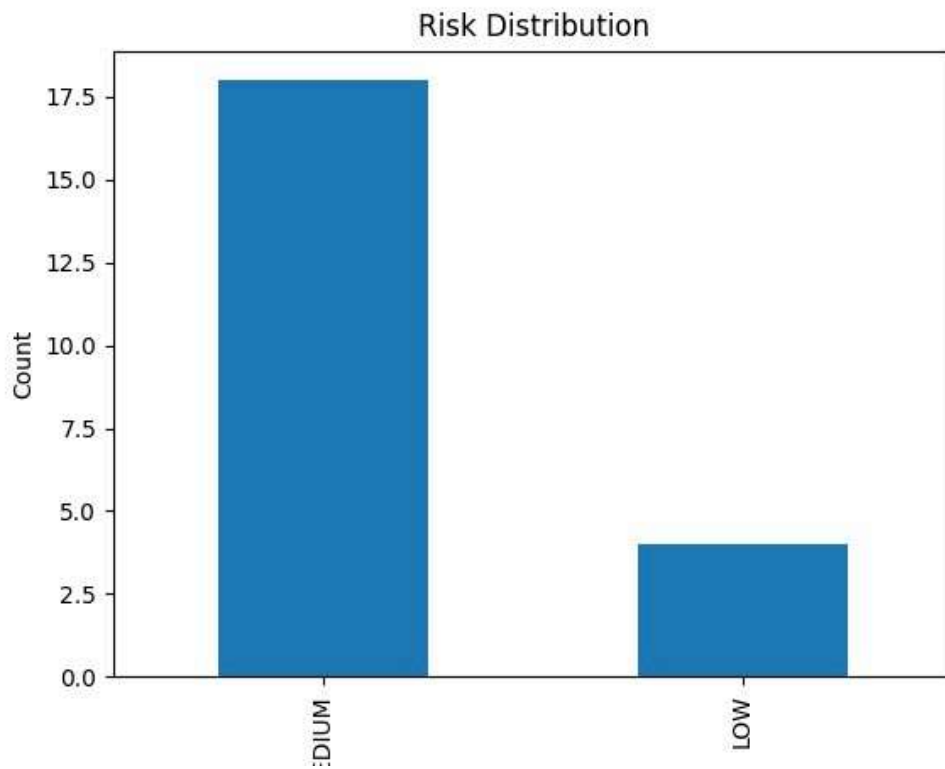
4. Risk Level Distribution (Bar Chart)

- Displays count of Low, Medium, and High risk levels
- Helps in understanding overall safety condition

Insight:

Higher number of high-risk values indicates potential danger.

Output :



Key Achievements :

- Successfully visualized real-time dam data
- Built an interactive dashboard for monitoring
- Clearly displayed prediction results
- Enabled easy understanding of risk conditions

Conclusion :

This milestone demonstrates how real-time data and machine learning predictions can be effectively visualized using dashboards. The dashboard helps in monitoring water levels, analyzing pressure changes, and identifying risk conditions, supporting better decision-making for dam safety.